MCO Wave Parameter Library

ADDENDUM REV B — Tweak 2: Extended Sweep to 210° with Flush Fairings

MCO-1A Piloted SAR Ornithopter · Supplements Rev A (Tweaks 3 & 4) · Issue date: 2026-06-02

1. Purpose

Rev A (Tweaks 3 and 4) introduced continuous variable sweep and dihedral bias for GA-speed cruise. Throughout that document the full-stow position was cited as 200° with a footnote stating that spring-loaded root fairings are required for operation above 68 kt. Tweak 2 was referenced but never formally specified.

This addendum formally documents Tweak 2: extending the maximum sweep angle from 200° to 210° and fitting spring-loaded clamshell root fairings so that stowed wings sit flush with the fuselage skin. Both changes are required before the Rev A variable-sweep modes operate correctly at their rated speeds.

---

2. Why 200° Stow is Insufficient Above 68 kt

2.1 Geometric drag at 200°

At 200° the wing has swept 20° past the straight-aft position (180°). The leading edge faces 20° into the oncoming flow. For a wing panel of chord 924 mm and mean semi-span 4.16 m, the normal force coefficient at this angle is $CD_n = 2 \times \sin^2(20°) = 0.234$. With eight wings this produces a stowed drag area of approximately 1.6 m² — larger than the fuselage drag area and the dominant drag source above 68 kt.

At 200° the wings also protrude slightly above the fuselage profile, creating local flow separation and interference drag that is difficult to fairing-treat without a structural seal.

2.2 What 210° provides

At 210° the wing has swept 30° past straight-aft. The chord now lies 10° past parallel with the fuselage centreline — the leading edge points very slightly inboard toward the fuselage surface. This allows the wing to nest into a channel moulded into the fuselage skin, sitting flush rather than proud. The angle to the oncoming flow drops from 20° to 10°, reducing CD_n from 0.234 to 0.060 — a four-fold reduction in stowed wing normal force before fairings are applied.

Stow angle	Angle to flow	CD_n	Drag area (8 wings)	Notes
180° (straight aft)	0°	0.000	~0.0 m²	Aerodynamically ideal; mechanically impractical — wing hangs para
200° (current spec)	20°	0.234	~1.60 m²	Hard stop provides positive lock. Wing protrudes above fuselage sk
210° (Tweak 2)	10°	0.060	~0.12 m²	Wing nests flush in fuselage channel. Stop loads reduced. Fairing s
210° + fairing closed	~0°	~0.010	~0.02 m²	Wing inside fairing. Drag contribution negligible.

2.3 Effect on the hard stop

The existing 200° hard stop resists the aerodynamic load trying to drive the wing from 200° back toward 180° (straight aft). At 200° this load is 38 N per wing at 240 km/h; the stop is designed for 115 N (3× safety factor). At 210° the same aerodynamic geometry applies but the load direction shifts: the flow now tries to drive the wing from 210° back toward 180° — the same direction, larger moment arm but smaller normal force due to lower CD_n. The net load at 210° is lower than at 200°. The existing 115 N hard stop specification is therefore unchanged — the stop simply relocates 10° further aft on the sweep axis.

---

3. Spring-Loaded Root Fairings

3.1 Function

A spring-loaded clamshell fairing at each wing root closes automatically when the wing reaches 210°. From outside the aircraft the wing station appears as a smooth continuation of the fuselage teardrop profile — no protrusion, no gap, no interference. The wing spar sits in a longitudinal channel in the fuselage side wall. The fairing shell forms the outer surface of that channel when closed.

When the FCC commands a wing to deploy (sweep forward from 210°), the leading edge of the wing pushes the fairing open against its spring. No actuator is required — the sweep servo provides the opening force. The fairing spring force is sized to hold the fairing closed at the maximum dive dynamic pressure but light enough that the sweep servo opens it without measurable speed penalty during deployment.

3.2 Geometry

Parameter	Value	Notes
Fairing material	2-ply CFRP (0°/90°)	Same layup as wing D-box skin
Fairing length	Chord + 50 mm overlap	Full chord coverage plus seal overlap
Fairing width (closed)	Wing thickness + 5 mm	Slight proud to ensure weather seal
Spring type	Torsion spring, stainless	At hinge line, both inboard and outboard
Spring closing force	8 N·m at closed position	Holds closed to 300 km/h dynamic pressure
Servo opening force	Sweep servo: 2.4 N·m available	×3 margin over spring — no delay on deploy
Hinge axis	Parallel to fuselage axis	Fairing opens laterally outboard
Mass per station	~420 g	Shell + spring + hinge hardware
Mass × 8 stations	~3.4 kg	Total added to MCO-1A MTOW
Gap seal	Silicone D-section bead	Bonded to fairing, compresses against fuselage

3.3 Deployment sequence (sweep out)

The following describes the opening sequence when the FCC commands a stowed wing to deploy — for example, during STOOP-4A pullout or transition from 210° sprint stow to W3+W4 cruise configuration.

Step	Event	Duration	FCC action
1	Sweep servo commanded from 210° toward 165°	t = 0	Sweep rate limited to 30°/s (normal) or 60°/s (stoop pullout)
2	Wing LE contacts fairing inner surface at ~205°	t ≈ 0.17 s	No action — spring begins compressing
3	Fairing opens: wing sweeps through 165°–120°	t ≈ 0.17–1.5 s	Normal sweep continues; fairing trails open
4	Wing reaches target deploy angle	t ≈ 1.5–3 s	Flap and pitch DoF re-enabled by FCC
5	Fairing remains open while wing deployed	—	FCC monitors sweep angle; fairing open if sweep < 200°
6	Re-stow: wing sweeps to 210°, fairing spring-closes	—	FCC confirms sweep = 210° before enabling high-speed sprint

---

4. Mechanism Changes Required

4.1 Sweep range extension: 200° → 210°

The ZOSKAY DS3225 sweep servo (MCO-1D) and its scaled equivalent for MCO-1A have a physical range of 270°. The 200° limit was a software parameter — SWEEP_MAX_DEGREES = 200 in the wave library. Tweak 2 changes this to 210. No servo hardware change is required on either vehicle.

The hard stop pin and collar on the sweep axle must be repositioned from 200° to 210°. On MCO-1D this is a machined aluminium collar with a press-fit pin — reposition requires removing the collar and rotating it 10° before re-pressing. On MCO-1A the stop is a separate CNC-machined component; the drawing requires a 10° change to the stop face angle.

Item	MCO-1D change	MCO-1A change
Software SWEEP_MAX_DEGREES	200 → 210 (one line)	200 → 210 (one line)
Hard stop position	Reposition collar 10°	Redraw stop face 10°
Sweep servo	No change	No change
Fairing hardware	Not fitted (drone)	New fabrication — see §3.2
Channel in fuselage	Not required (drone)	Moulded into fuselage layup

Note: MCO-1D does not require fairings. The drone flies without them and the stow drag at 200° is acceptable at drone speeds (max 95 km/h stoop). The 210° extension on the drone provides clearance for the revised stow geometry and validates the servo range before MCO-1A build.

4.2 Revised stow order (unchanged from base spec)

The aft-to-forward stow cascade (W4 → W3 → W2 → W1, 0.2 s offset) is unchanged. With the 10° extension to 210° the inter-wing clearance during stow is marginally improved — the 10° extra puts each wing slightly deeper into the fuselage channel before the next wing begins its stow sweep.

---

5. Wave Library Parameter Changes

5.1 Single parameter change (Tweak 2 core)

```
# TWEAK 2 — Extended stow to 210° # Change in wave_library.py, MCO_FCC module # Before: SWEEP_MAX_DEGREES = 200 # hard stop position STOW_ANGLE = 200 # full stow target # After (Tweak 2): SWEEP_MAX_DEGREES = 210 # hard stop repositioned to 210° STOW_ANGLE = 210 # full stow target # All references to stow angle in STOOP sequence, sweep schedule, # and variable sweep modes (Rev A) automatically use STOW_ANGLE. # No other parameter changes required.
```

5.2 Fairing status flag (MCO-1A only)

```
# TWEAK 2 — Fairing configuration flag (MCO-1A only) # Enables high-speed sprint modes that require fairings FAIRINGS_FITTED = True # Set False on MCO-1D (no fairings) # FCC uses this flag to gate sprint modes: # If FAIRINGS_FITTED is False, max speed limited to 68 kt # (W3+W4 cruise without fairings — validated physics). # If FAIRINGS_FITTED is True, full Rev A sweep schedule active # and 97–120 kt sprint modes available.
```

---

6. Revised Stow Drag — Impact on Speed Envelope

The table below shows how stow drag evolves across the three configurations. The stoop terminal velocity figures are for the MCO-1A with enclosed teardrop fuselage (CD·S = 0.12 m²). Compressibility is not modelled — actual terminal velocity will be limited by structural and pilot tolerances before the aerodynamic limit is reached.

Configuration	Stowed wing drag area	Fuselage drag area	Total drag area	Stoop terminal velocity	Status
Open cockpit, 200° stow (original baseline)	1.60 m²	0.42 m²	2.02 m²	~43 kt	Historical ref only
Enclosed cockpit, 200° stow (no fairings)	1.60 m²	0.12 m²	1.72 m²	~49 kt	Interim — fairings not fitted
Enclosed cockpit, 210° stow (no fairings)	0.41 m²	0.12 m²	0.53 m²	~88 kt	Tweak 2 sweep only
Enclosed cockpit, 210° stow + fairings closed (Tweak 2 full)	~0.02 m²	0.12 m²	~0.14 m²	~156 kt*	Target — full Tweak 2

* 156 kt is the aerodynamic terminal velocity. Structural and pilot load limits apply — see STOOP-2A specification for operational ceiling.

---

7. Interaction with Addendum Rev A (Tweaks 3 and 4)

Tweak 2 is a prerequisite for the full Rev A speed envelope. The Rev A sweep schedule uses 200°* with a footnote — that asterisk refers to this addendum. The corrected sweep schedule with Tweak 2 applied:

Speed lever	Airspeed	W1	W2	W3	W4	Fairings	Mode
Detent / 0–30%	<32 kt	60°	60°	60°	60°	Open	Full deploy — soar/slow cruise
30–50%	32–46 kt	120°	60°	60°	60°	Open	W1 sweeping
50–65%	46–68 kt	210°	210°	60°	60°	Closed†	W3+W4 cruise — swift
65–80%	68–97 kt	210°	210°	135°	135°	Closed†	All wings delta — GA approach
80–95%	97–120 kt	210°	210°	135°	135°	Closed†	GA sprint
95–100%	>120 kt	210°	210°	165°	210°	Closed†	Pre-stoop
STOOP switch	—	210°	210°	210°	210°	Closed†	Full stow — STOOP-2A

† Fairings close automatically when wing reaches 210°. FCC gates 68+ kt modes on FAIRINGS_FITTED = True. MCO-1D (no fairings) limited to 68 kt regardless of lever position.

---

8. Open Items

#	Item	Dependency	Priority
O-1	Reposition hard stop collar on MCO-1D from 200° to 210°	Workshop access	High — required before testing Rev A sweep schedule
O-2	Update SWEEP_MAX_DEGREES and STOW_ANGLE in wave_library.py	None	High — one-line change, no risk
O-3	Fabricate 8× fairing assemblies for MCO-1A	MCO-1A fuselage mould	Medium — required before 68+ kt sprint modes
O-4	Measure spring opening force at sweep servo (confirm 3× margin)	Fairing prototype	Medium
O-5	MCO-1D flight test: verify 210° stow geometry and stow order clearance	O-1, O-2 complete	High — gates MCO-1A fairing design
O-6	Update all wave library tables referencing '200° stow' to '210° stow'	None	Low — documentation only, parameters already corrected ab

---

MCO Wave Parameter Library — Addendum Rev B. Tweak 2: sweep range extended 200° → 210°, spring-loaded root fairings. Hard stop repositioned on sweep axle. Two software parameter changes: SWEEP_MAX_DEGREES = 210, STOW_ANGLE = 210. Prerequisite for full Rev A (Tweaks 3 and 4) sprint speed envelope. MCO-1D requires hard stop reposition only — no fairings.